

Differential Evolution Task: Robot Basketball Shot Optimizer

Course: Evolutionary Algorithms

Task Title

Using Differential Evolution to Find the Best Basketball Shot

Problem Statement

A robot is trying to throw a basketball into a hoop. The robot can control two variables:

$$X = [v, \theta]$$

where

v = initial speed of the ball in m/s

θ = launch angle in degrees

The robot must choose suitable values of v and θ so that the ball reaches the hoop accurately while also using reasonable shooting power.

Given Data

The robot releases the ball from the height

$$h_0 = 1.2 \text{ m}$$

The hoop is located at the horizontal distance

$$D = 8 \text{ m}$$

The hoop height is

$$H = 3.05 \text{ m}$$

The acceleration due to gravity is

$$g = 9.81 \text{ m/s}^2$$

There is also a defender standing halfway between the robot and the hoop. The ball should pass above the defender.

$$x_d = 4 \text{ m}$$

$$H_d = 2.4 \text{ m}$$

Decision Variable Bounds

The allowed ranges for the decision variables are

$$5 \leq v \leq 15$$

$$25^\circ \leq \theta \leq 75^\circ$$

Projectile Equation

Before using the launch angle in the equation, convert it from degrees to radians:

$$\theta_{\text{rad}} = \frac{\pi}{180} \theta$$

The height of the ball at a horizontal distance x is given by

$$y(x) = h_0 + x \tan(\theta_{\text{rad}}) - \frac{gx^2}{2v^2 \cos^2(\theta_{\text{rad}})}$$

The ball height at the hoop is

$$y(D)$$

The ball height at the defender is

$$y(x_d)$$

Objective Function

The main goal is to make the ball height at the hoop close to the actual hoop height. Therefore, the hoop-height error is

$$J_1 = (y(D) - H)^2$$

To avoid unnecessarily powerful shots, add a small speed penalty:

$$J_2 = 0.002v^2$$

If the ball does not pass above the defender, apply a defender penalty:

$$J_3 = 20 \times \max(0, H_d - y(x_d))^2$$

The final objective function is

$$J = J_1 + J_2 + J_3$$

The goal is

Minimize J

Part A: DE Calculation by Hand

Use a small population of your own choice. The population size must be at least

$$N \geq 4$$

Each individual should contain two variables:

$$X_i = [v_i, \theta_i]$$

Use the following Differential Evolution settings for hand calculation:

$$F = 0.5$$

$$CR = 0.7$$

Use the **DE/rand/1/bin** strategy.

Step 1: Mutation

For a target vector X_i , choose three different vectors X_{r_1} , X_{r_2} , and X_{r_3} from the population such that

$$r_1 \neq r_2 \neq r_3 \neq i$$

The mutant vector is generated as

$$V_i = X_{r_1} + F(X_{r_2} - X_{r_3})$$

Step 2: Crossover

The trial vector is generated using binomial crossover:

$$U_{ij} = \begin{cases} V_{ij}, & \text{if } rand \leq CR \text{ or } j = j_{\text{rand}}, \\ X_{ij}, & \text{otherwise.} \end{cases}$$

Here, j is the variable index. Since this problem has two variables, $j = 1$ corresponds to speed v and $j = 2$ corresponds to launch angle θ .

The term j_{rand} ensures that at least one component of the trial vector comes from the mutant vector.

Step 3: Greedy Selection

For a minimization problem, greedy selection is applied as

$$X_i^{\text{new}} = \begin{cases} U_i, & \text{if } J(U_i) < J(X_i), \\ X_i, & \text{otherwise.} \end{cases}$$

Hand Calculation Requirement

For hand calculation, perform the DE steps for at least **one target vector**. Clearly show:

1. Initial population.
2. Fitness value of each individual.
3. Selected indices r_1 , r_2 , and r_3 .
4. Mutant vector calculation.
5. Trial vector calculation.

6. Fitness comparison between target vector and trial vector.
7. Greedy selection decision.

You can choose your own random numbers but clearly mention them.

Part B: Python Implementation

Write a Python program to solve the same robot basketball shot problem using Differential Evolution.

Students may use

`scipy.optimize.differential_evolution`

or they may implement Differential Evolution manually.

You are free to choose

population size

and

number of generations

However, your selected values must be clearly mentioned in your report.

Required Output

Your program should print the following values:

Quantity	Value
Selected population size	
Selected number of generations	
Best speed v	
Best angle θ	
Ball height at hoop $y(D)$	
Ball height at defender $y(x_d)$	
Final objective value J	

Also plot:

1. Best objective value versus generation.
2. Best speed and best angle versus generation.

Questions to Answer

Answer the following questions in your submission:

1. What does one individual represent in this problem?
2. What is the target vector in Differential Evolution?
3. What is the mutant vector?
4. What is the trial vector?
5. Why is the defender penalty added?
6. What is the role of the differential weight F ?
7. What is the role of the crossover rate CR ?
8. Did your best shot pass above the defender?
9. Was the final shot accurate near the hoop height?
10. Why may different students obtain different final answers?

Submission Requirements

Submit the following:

1. Hand calculation for one target vector.
2. Python code.
3. Screenshot or copy of program output.
4. Convergence plots.
5. Final best solution table.
6. Answers to the questions.